陈信均 12311125

Homework for Chap 8

With the effective mass approximation, do the following calculations:

1. Suppose the bandgap of GaAs at T=300 K is 1.42 eV and the electron effective mass is 0.062e$_0$, calculate the density of states at energy of 1.52 eV and the effective density of states for nondegenerate electron gas in three dimensional (3D) GaAs.
2. For a GaAs quantum well with the well width of 3 nm, suppose the barrier height is infinite, calculate the first quantized energy E_1 for the 2-dimensional (2D) electrons and the corresponding electron density of states at this quantized energy E_1.
3. Calculate the electron density of states for 3D electron gas at the energy E_1, obtained in Q2, and compare the value with the density of states for 2D electrons at E_1.

1. $g(E) = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{\frac{3}{2}}\sqrt{E-E_C}$

$= \frac{1}{2\pi^2}\left(\frac{0.062m_e\times 2}{\hbar^2}\right)^{\frac{3}{2}}\sqrt{E-E_C} = 2.08\times 10^{44}\, m^{-3}J^{-1} = 3.33\times 10^{25}\, m^{-3}eV^{-1}$

If the spin is considered. $g(E) = 2\times 3.3\times 10^{25} m^{-3}eV^{-1} = 6.6\times 10^{25} m^{-3}eV^{-1}$

Effective density states:

$N_C = 2\left(\frac{2\pi m^* k_B T}{h^2}\right)^{\frac{3}{2}} = 3.87\times 10^{23}\, m^{-3}$

2. $E_1 = \frac{\hbar^2\pi^2}{2m^*L^2} \Rightarrow E_1 = 1.080\times 10^{-19} J = 0.674\ eV$

$g_{2D} = \frac{m^*}{\pi\hbar^2} = 1.62\times 10^{36}\, m^{-2}J^{-1}$

$= 2.6\times 10^{17}\, m^{-2}eV^{-1}$

If the spin is considered, $g_{2D} = 5.2\times 10^{17} m^{-2} eV^{-1}$

3. $g_{3D}(E_1) = \frac{1}{2\pi^2}\left(\frac{2m^*}{\hbar^2}\right)^{\frac{3}{2}}\sqrt{E_1} = 5.38\times 10^{64}\, m^{-3}J^{-1}$

If the spin is considered, $g_{3D} = 1.7\times 10^{26} m^{-3}eV^{-1} = 8.63\times 10^{25}\, m^{-3}eV^{-1}$

$\boxed{g_{3D} > g_{2D}}$ $\qquad g_{3D} = \frac{g_{2D}}{L}$